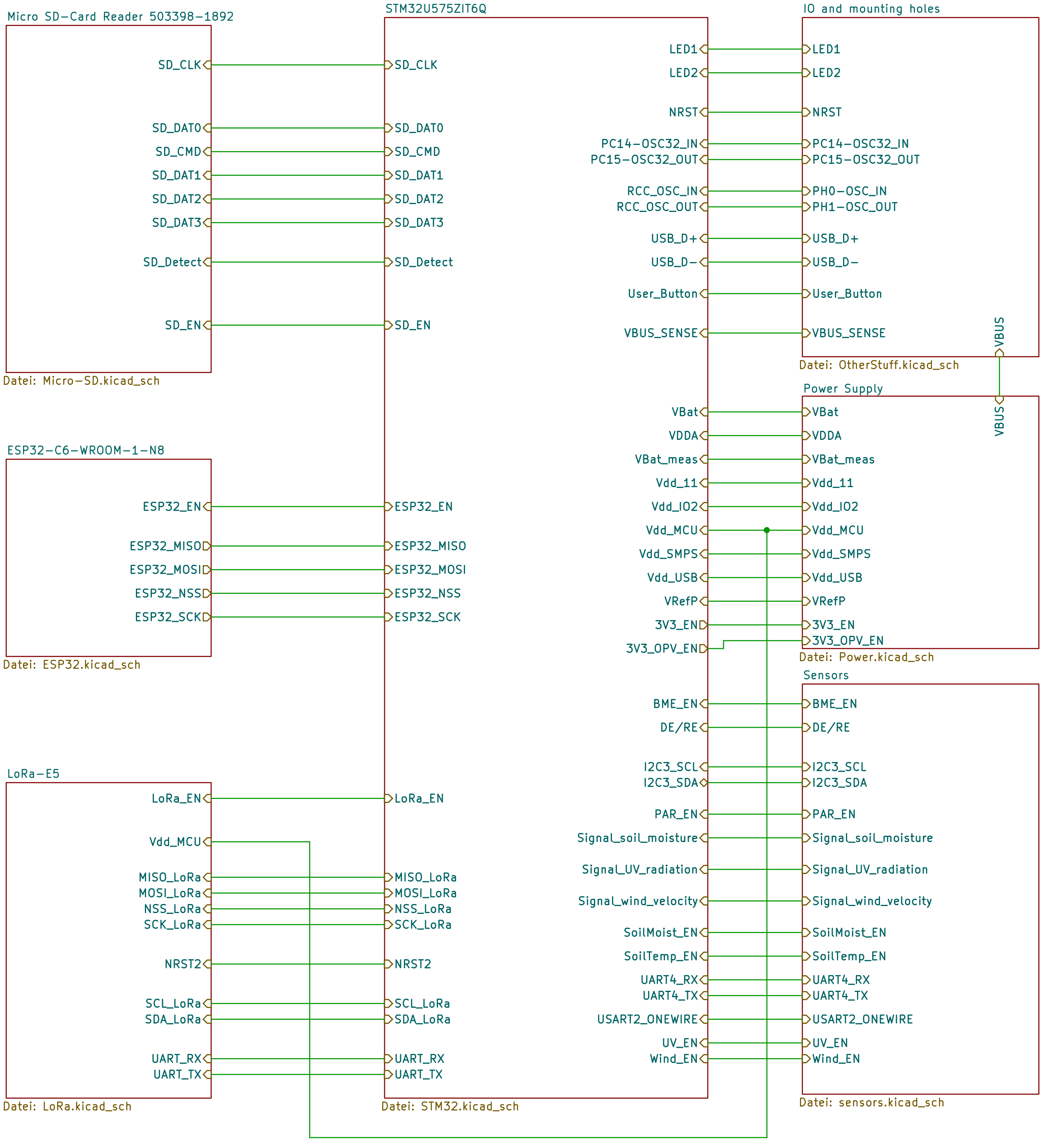
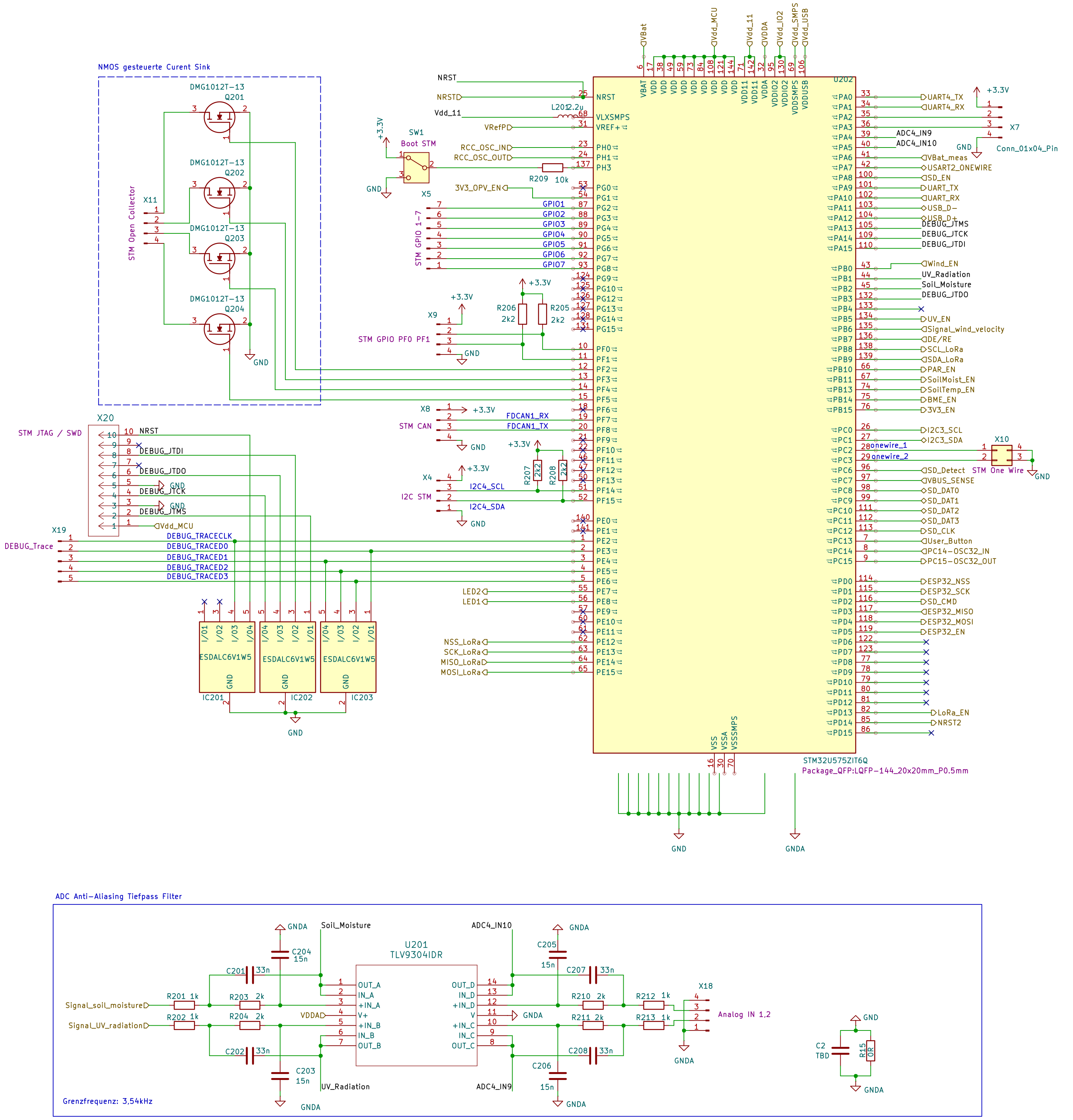


Hochschule Karlsruhe

Sheet: /
File: uC_Agri-PV.kicad_sch

Title: µC Agri-PV

Size: A4	Date: 2026-01-08	**Rev: 2.0**
KiCad E.D.A. 9.0.5		Id: 1/8



Hochschule Karlsruhe

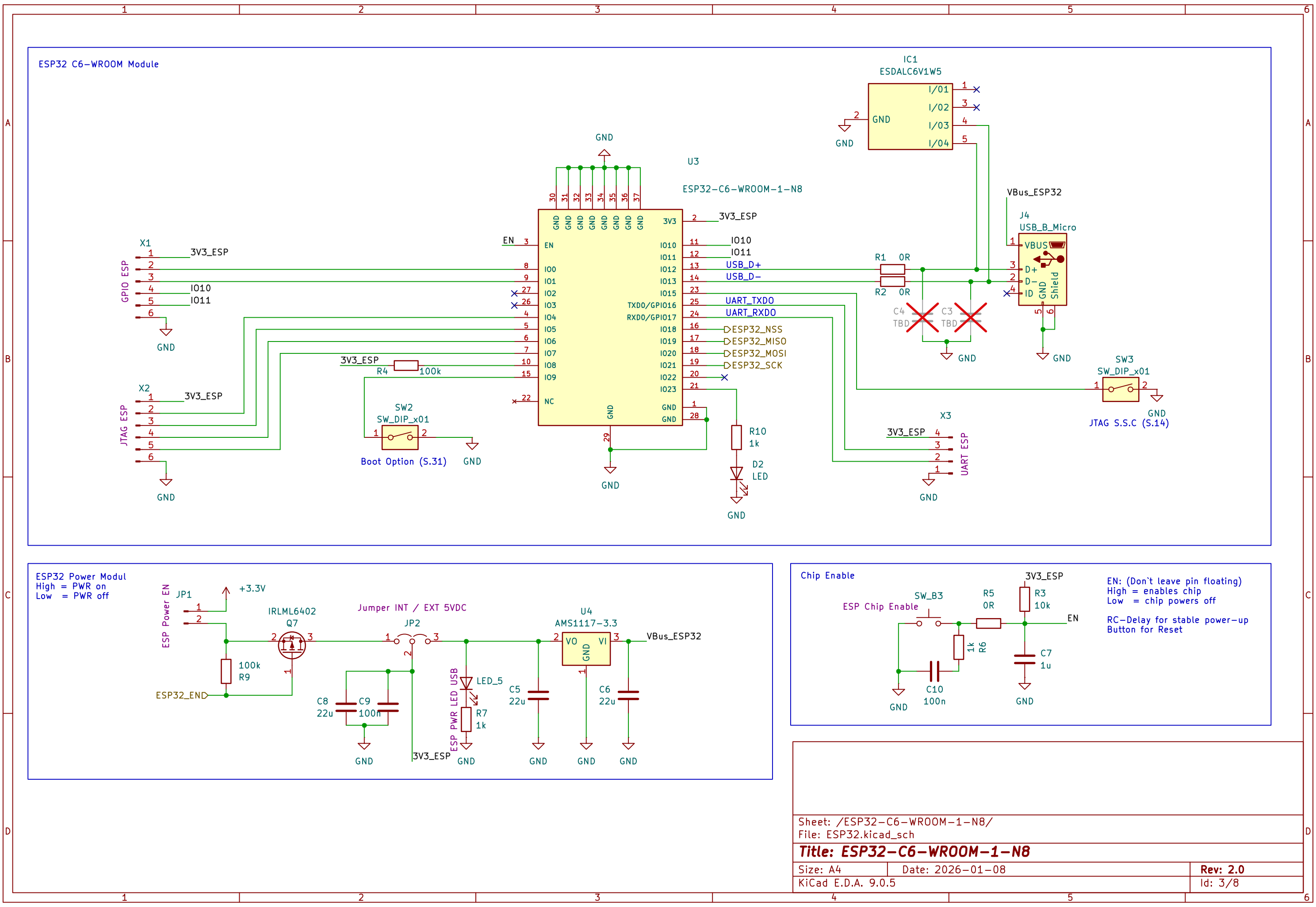
Sheet: /STM32U575ZIT6Q/

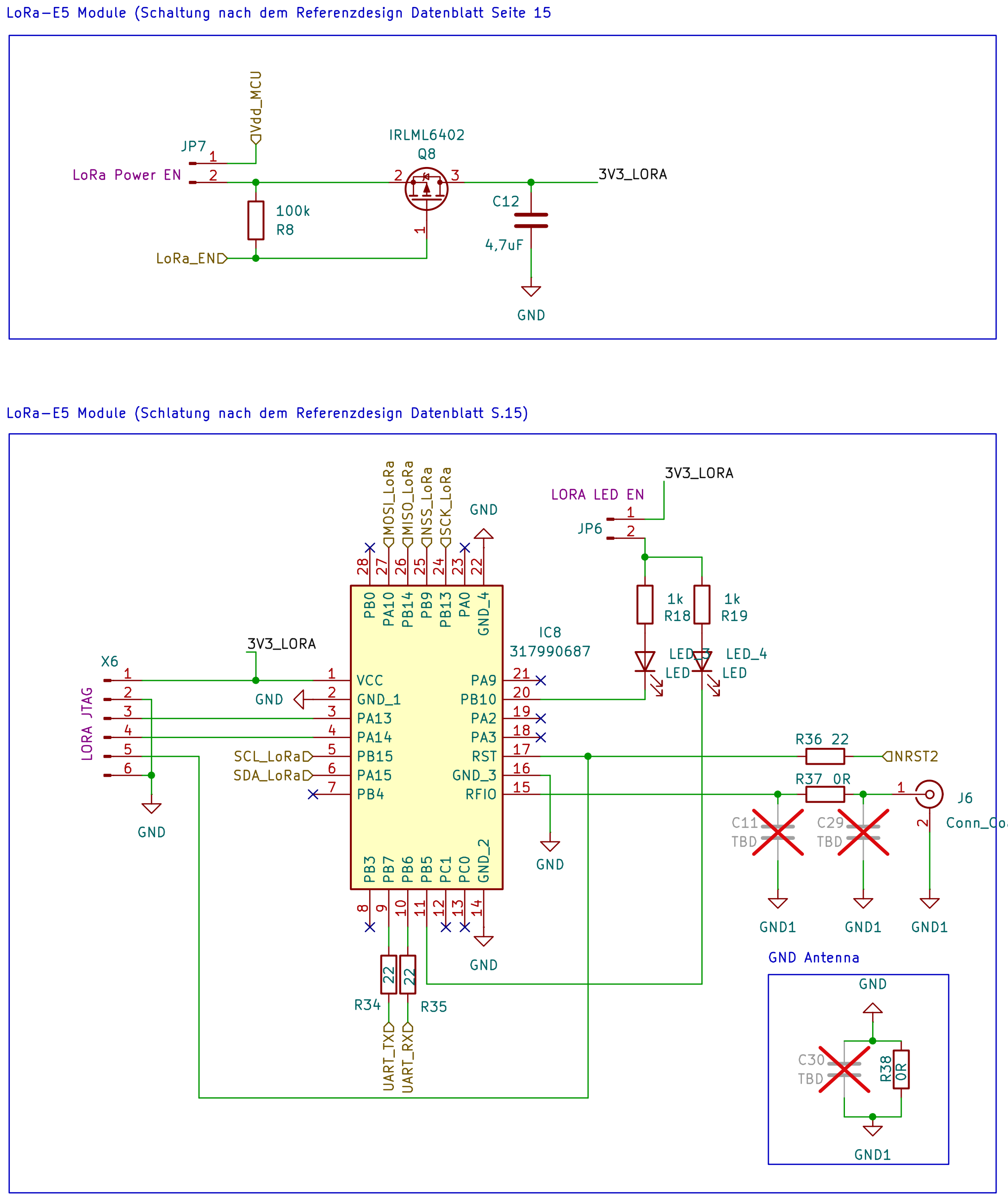
File: STM32.kicad_sch

Title: µC Agri-PV

Size: A3 | Date: 2026-01-08 | Rev: 2.0

KiCad E.D.A. 9.0.5 | Id: 2/8


ESP32 C6-WROOM Module
IC1
ESDALC6V1W5
GND
I/O1
I/O2
I/O3
I/O4
U3
ESP32-C6-WROOM-1-N8
3V3_ESP
EN
IO10
IO11
USB_D+
USB_D-
UART_TXDO
UART_RXDO
ESP32_NSS
ESP32_MISO
ESP32_MOSI
ESP32_SCK
TXD0/GPIO16
RXD0/GPIO17
NC
X1
GPIO ESP
X2
JTAG ESP
X3
UART ESP
R4
100k
SW2
SW_DIP_x01
Boot Option (S.31)
R10
1k
D2
LED
R1 0R
R2 0R
C4 TBD
C3 TBD
VBus_ESP32
J4
USB_B_Micro
VBUS
D+
D-
ID
Shield
SW3
SW_DIP_x01
JTAG S.S.C (S.14)
GND
ESP32 Power Modul
High = PWR on
Low = PWR off
ESP Power EN
JP1
+3.3V
IRLML6402
Q7
100k
R9
ESP32_END
Jumper INT / EXT 5VDC
JP2
C8 22u
C9 100n
ESP PWR LED USB
LED_5
R7 1k
C5 22u
U4
AMS1117-3.3
VO
VI
C6 22u
Chip Enable
ESP Chip Enable
SW_B3
R5 0R
R3 10k
R6 1k
C10 100n
C7 1u
EN: (Don't leave pin floating)
High = enables chip
Low = chip powers off
RC-Delay for stable power-up
Button for Reset
Sheet: /ESP32-C6-WROOM-1-N8/
File: ESP32.kicad_sch
Title: ESP32-C6-WROOM-1-N8
Size: A4
Date: 2026-01-08
KiCad E.D.A. 9.0.5
Rev: 2.0
Id: 3/8

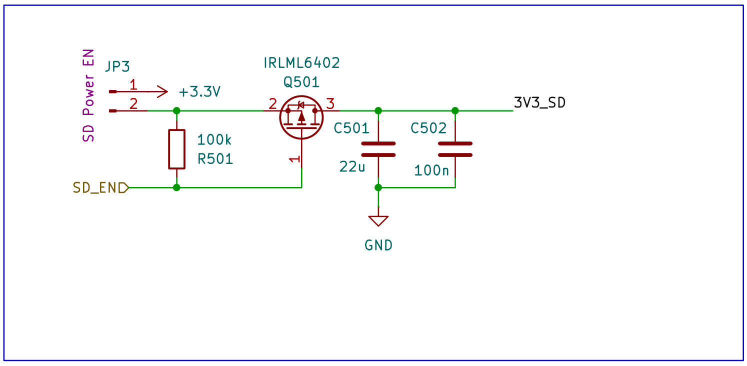


Sheet: /LoRa-E5/
File: LoRa.kicad_sch

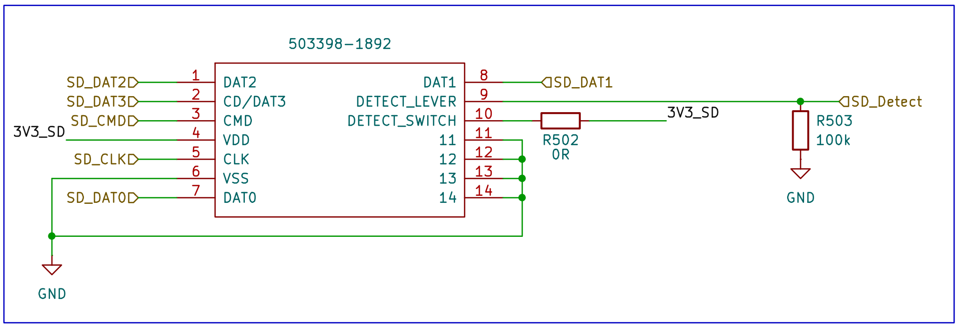
Title: LoRa RFM95W-868S2

Size: A4	Date: 2026-01-08	Rev: 2.0
KiCad E.D.A. 9.0.5		Id: 4/8

Micro−SD Power



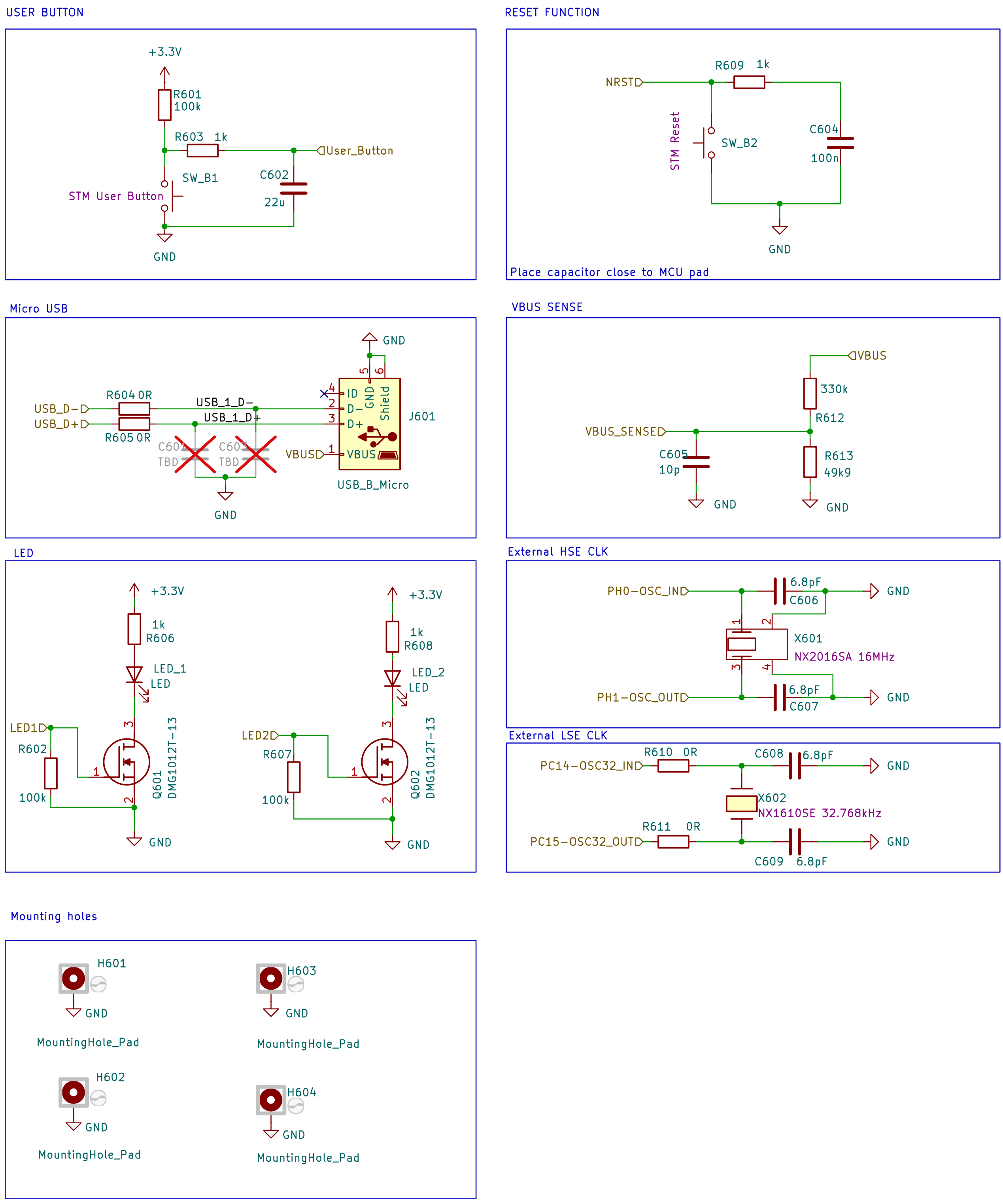
Micro−SD IO



Sheet: /Micro SD−Card Reader 503398−1892/
File: Micro−SD.kicad_sch

Title: Micro−SD Reader

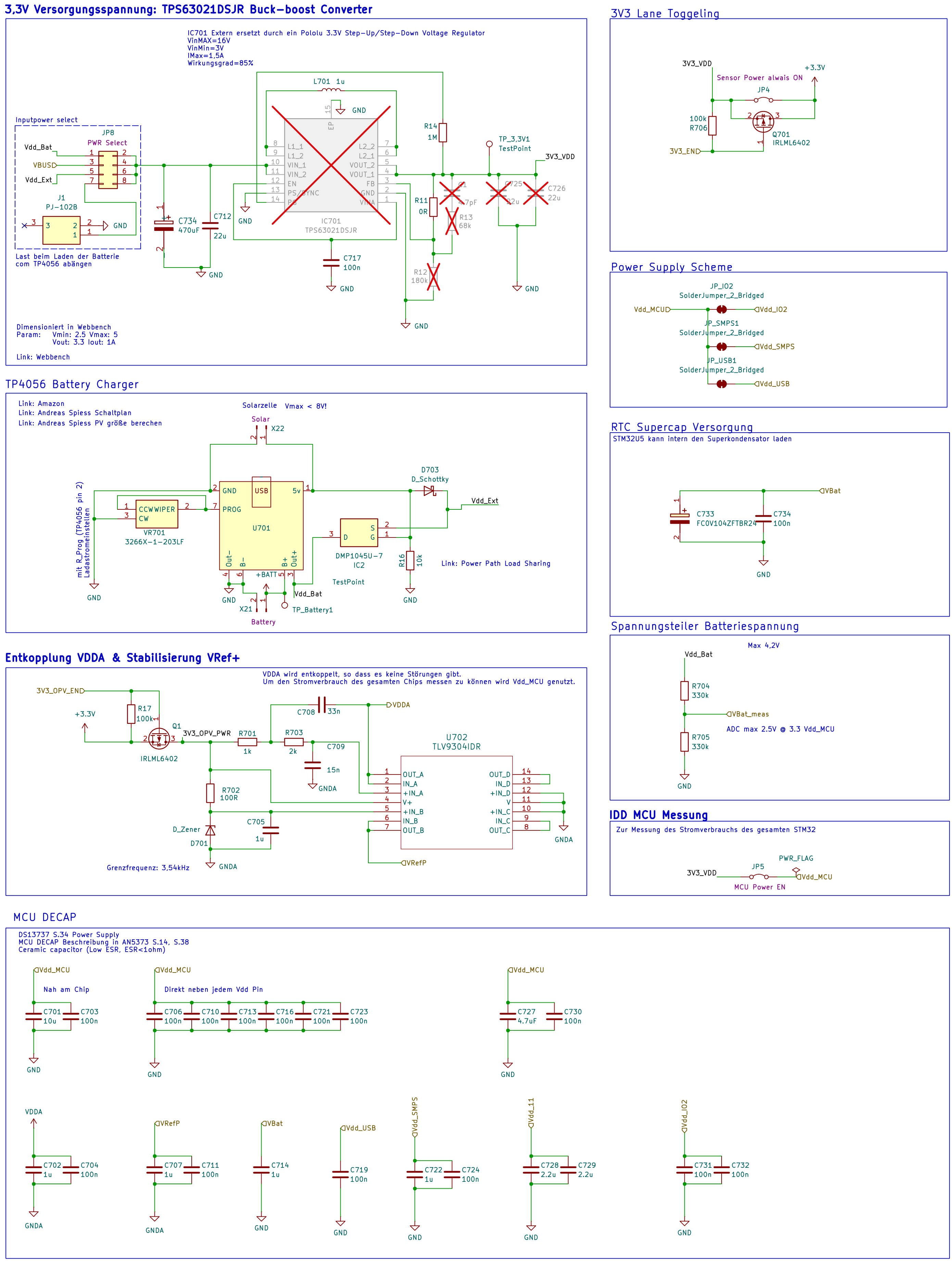
Size: A4 | Date: 2026−01−08 | **Rev: 2.0**
KiCad E.D.A. 9.0.5 | Id: 5/8



Sheet: /IO and mounting holes/
File: OtherStuff.kicad_sch

Title: GPIO and other stuff

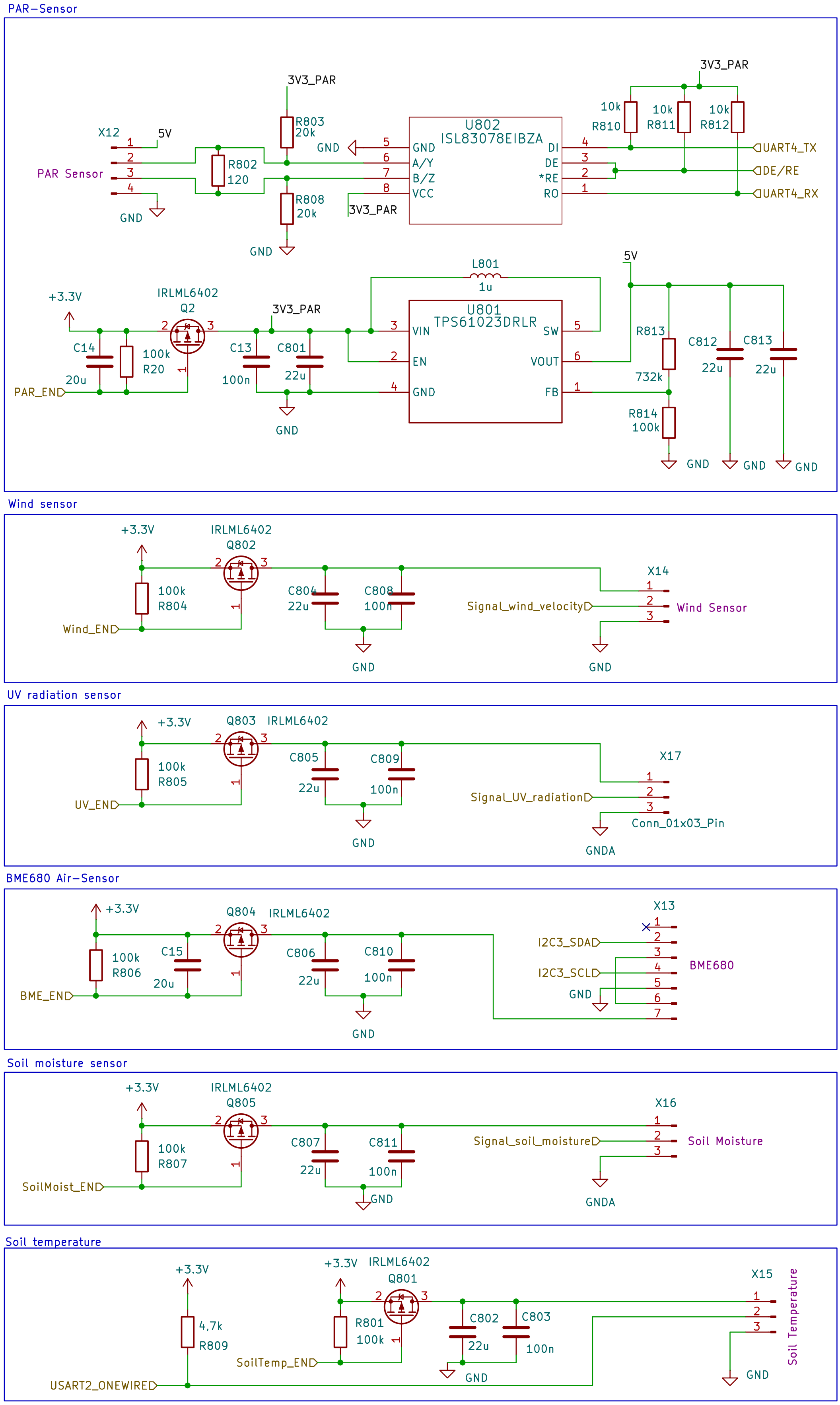
Size: A4	Date: 2026-01-08	Rev: 2.0
KiCad E.D.A. 9.0.5		Id: 6/8



Sheet: /Power Supply/
File: Power.kicad_sch

Title: Power Supply

Size: A3	Date: 2026-01-08	Rev: 2.1
KiCad E.D.A. 9.0.5		Id: 7/8



Sheet: /Sensors/

File: sensors.kicad_sch

Title: Used Sensors

Size: A4	Date: 2026−01−08	**Rev: 2.0**
KiCad E.D.A. 9.0.5		Id: 8/8